Lost in Legal Language: The Failure of Zero-Shot AI Text Detectors

Abhilash Shankarampeta
UCSD

Ananay Gupta
UCSD

Hemanth Bodala
UCSD

Rohan Surana
UCSD

Sarthak Kala
UCSD

Vignesh Palaniappan
UCSD

ABSTRACT

The growing use of large language models (LLMs) in legal drafting raises concerns about hallucinated citations, factual inaccuracies, and unintended disclosure of sensitive information. In high-stakes domains like law, such errors can have serious consequences, as seen in the *Mata v. Avianca* case. Yet, reliable tools for verifying citations, detecting hallucinations, and flagging confidential content in AI-generated legal text remain lacking. To explore whether AI-written text in the legal domain exhibits detectable statistical signatures, we evaluate three modern detection methods: (1) BiScope, (2) Binoculars, and (3) OOD-DeepSVDD on both real and high-quality synthetic legal data. Results show that zero-shot detectors perform poorly in this domain: BiScope misses 99% of AI-generated text, Binoculars over-flags human-written documents, and OOD-DeepSVDD struggles due to high-quality synthetic samples. However, fine-tuning BiScope with domain-specific data improves recall from 0.57% to 66%. Our findings underscore the need for domain-adaptive solutions for trustworthy AI detection in legal contexts. Code and data are available at: https://github.com/rohan2810/Lost-in-Legal-Language

1 INTRODUCTION

In *Mata v. Avianca* (S.D.N.Y., 2023), attorneys submitted non-existent cases and were sanctioned, drawing early attention to the risks of unverified AI-generated content in litigation.[13] Recent incidents show that these problems are no longer isolated. In *Noland v. Land of the Free* (Cal. Ct. App. 2025), a California attorney was sanctioned $10,000 after an appellate brief relied on fake citations generated by an LLM.[1] Likewise, in *Wadsworth v. Walmart* (D. Wyo. 2025), three Morgan & Morgan attorneys were sanctioned for submitting a motion containing eight non-existent, AI-generated cases.[2]

A recent cataloging effort reports nearly one hundred such AI-related "hallucination" incidents across courts in the United States and abroad, suggesting a growing systemic risk rather than a handful of outliers.[3]

Currently, there is no reliable system to: (a) verify cited authorities, (b) detect hallucinated or inconsistent content, or (c) flag confidential or privileged data in AI-assisted legal drafts. These gaps hinder lawyers' ability to meet professional standards.

To explore whether AI-written text in the legal domain exhibits detectable statistical signatures, we evaluate three modern detection methods: BiScope, Binoculars, and OOD-DeepSVDD on both real and high-quality synthetic samples. Our focus is to assess whether these detectors, originally developed for general text, can reliably distinguish human-written legal prose from LLM-generated content and inform safer AI-assisted drafting practices.

2 RELATED WORKS

The detection of LLM-generated text has become increasingly critical as language models achieve near-human text quality.

2.1 Zero-Shot Detection Methods

DetectGPT [11] introduced a zero-shot method based on the observation that LLM-generated text lies in regions of negative curvature in the model's log probability function, but its reliance on 100 model calls per passage limits scalability. Fast-DetectGPT [1] overcomes this with conditional probability curvature, achieving a 340× speedup while retaining strong accuracy. Binoculars [7] offers state-of-the-art zero-shot detection via dual-model perplexity comparison, identifying over 90% of ChatGPT outputs at just 0.01% false positive rate. Other methods include GLTR [5], which combines token-level log probabilities, ranks, and entropy for interpretability; DetectLLM [14], which uses log-rank features; and DNA-GPT [21], which applies divergent N-gram analysis by truncating and regenerating text to measure divergence.

2.2 Supervised Detection Approaches

Ghostbuster [17] achieved state-of-the-art supervised detection using probabilities from multiple weaker models with structured feature selection. The method's black-box operation and strong cross-model generalization make it practical for detecting responses of proprietary models, such as Claude and GPT. StyloAI [12] demonstrated effective feature-based detection using 31 stylometric features, including lexical diversity, syntactic complexity, and readability metrics, achieving state-of-the-art accuracy while providing interpretable features.

2.3 Domain-Specific Considerations

DetectRL [20] demonstrated that state-of-the-art detectors perform poorly in realistic adversarial conditions, with performance varying by text length, domain, and attack method. MAGE [9] showed the best detector, achieving only 84.12%

[1] See, e.g., California Attorney Fined $10k for Filing an Appeal With Fake Legal Citations Generated by AI, *U.S. News & World Report*, Sept. 22, 2025.

[2] See, e.g., Morgan & Morgan Lawyers Fined for "Hallucinated" AI Citations, *Bloomberg Law*, Feb. 24, 2025.

[3] See D. Charlotin, Hallucinations, https://www.damiencharlotin.com/hallucinations/ (last visited [date]).

accuracy on out-of-domain texts from new LLMs. Cross-domain generalization remains challenging, with detectors trained on specific LLMs failing on newer models [18, 19].

Colombo et al. [4] introduced SaulLM-7B, a legal-domain LLM trained on 30B+ legal tokens, demonstrating the emergence of specialized legal language models that detection methods must address.

3 METHODOLOGY

3.1 Dataset Generation

We utilized the EURLEX57K dataset [3], accessible via HuggingFace as `jonathanli/eurlex`, which comprises 57,000 legislative documents in English from the EUR-Lex portal (https://eur-lex.europa.eu). The dataset was curated for multi-label text classification and few-shot and zero-shot learning research. Each document has an average length of 727 words and is structured into three major sections: the header (including title and legal body), recitals (legal background references), and the main body (organized in articles). All documents are annotated by the Publications Office of the EU with EUROVOC concepts for multi-label classification tasks. From this dataset, we combined the `train`, `validation`, and `test` splits, shuffled with a fixed random seed (42), and sampled **10,000** documents for synthetic generation.

3.1.1 Synthetic Data Generation Model. For synthetic legal document generation, we used **four** instruction-tuned large language models: *Qwen3-8B [16]*, *Llama-3.1-8B-Instruct [6]*, *Mistral-7B-Instruct-v0.3 [8]*, and *gemma-3-4b-it [15]*. We generated **2,500** synthetic documents per model (**10,000** total). All models were run in `bfloat16` precision using HuggingFace Transformers. For Qwen3-family models, we disabled the optional "thinking" behavior to avoid emitting intermediate reasoning tokens, since the task is paraphrasing rather than multi-step reasoning. Generation uses batched decoding with batch size 32, maximum new tokens 2048, temperature = 0.7, and top-p = 0.9.

3.1.2 Data Generation Strategy. Each of the 10,000 sampled documents is assigned a boolean flag indicating whether case numbers are preserved. We enforce a 50/50 split and then randomly shuffle assignments so the preservation setting is distributed across the dataset. This choice is motivated by our downstream task of distinguishing real documents from generated documents. If all synthetic samples always modified case numbers, then case-number changes could become a dominant and potentially misleading cue for detection. Using a mixture of preserved and modified citations reduces this reliance and helps evaluate detection performance under a more realistic range of citation behavior. This design also accounts for the possibility that some models may have memorized common legal citations during pretraining, which could influence both generation and detection when case numbers are preserved.

3.1.3 Prompting Strategy. We designed a structured prompt to guide the model in generating synthetic legal text while preserving the formal legal language style and document structure. Two prompt variants were employed to create diversity in the synthetic dataset: 50% of samples used a prompt that changes all identifying information, including case numbers, while 50% preserved original case numbers to maintain structural fidelity to legal citations. In early experiments, we attempted to enforce case-number preservation using a single instruction-only prompt (similar to the full anonymization prompt), but observed failures where the model failed to follow the instructions. **To improve instruction-following, we switched to an in-context learning formulation with two illustrative examples for the preservation setting, which substantially improved adherence to the case-number constraint.**

The full prompt templates are provided in the Appendix (Variant 1 in Appendix 1 and Variant 2 in Appendix 2).

3.2 BiScope: Bi-directional Cross-Entropy Detection

We implemented BiScope [10], a detection method that exploits bi-directional cross-entropy patterns to distinguish AI-generated from human-written text. Unlike traditional approaches that rely solely on forward language modeling probability, BiScope computes both forward and backward cross-entropy losses under the hypothesis that AI-generated text exhibits asymmetric patterns: lower forward loss due to optimized next-token prediction, but higher backward loss because language models are not trained to reconstruct previous tokens. Human-written text, in contrast, maintains natural bidirectional coherence.

Our implementation uses two surrogate language models—GPT-2 (124M parameters) and DistilGPT-2 (82M parameters)—as frozen feature extractors. For each document, we compute token-level cross-entropy losses in both directions:

- **Forward loss**: For each token position i, we calculate $\mathrm{CE}_{\mathrm{forward}} i = -\log Pt_{i1} | t_1, ..., t_i$
- **Backward loss**: For each token position i, we calculate $\mathrm{CE}_{\mathrm{backward}} i = -\log Pt_{i-1} | t_1, ..., t_i$

From each loss sequence, we extract five statistical features: mean, standard deviation, minimum, maximum, and median. With two surrogate models and two loss directions per model, this yields a 20-dimensional feature vector per document (5 statistics × 2 directions × 2 models).

These features are classified using a Random Forest ensemble [2] with 100 decision trees and a maximum depth of 10. Critically, while the surrogate language models require no training or fine-tuning, the classifier must be trained on labeled data from the target domain. This makes BiScope a weakly supervised rather than truly zero-shot approach.

3.2.1 Baseline: Detection Without Training. To assess the necessity of supervised learning, we evaluated BiScope features using a simple threshold-based approach without training the Random Forest classifier. We computed the difference between average backward and forward losses across both surrogate models and selected an optimal threshold via ROC curve analysis on the full dataset. This represents the best achievable performance using raw BiScope features alone.

3.3 Binoculars

Binoculars [7] is a zero-shot detection method that identifies machine-generated text by comparing perplexity measurements across two language models. The method computes two metrics: (1) the perplexity of the text according to an "observer" model M_1, which measures how surprising the text is, and (2) the cross-perplexity between two models M_1 and M_2, which measures how surprising M_2's next-token predictions are to M_1.

The Binoculars score is defined as the ratio:

$$Bs = \frac{\log \mathrm{PPL}_{M_1} s}{\log \text{X-PPL}_{M_1,M_2} s}. \tag{1}$$

where

$$\log \mathrm{PPL}_{M_1} s = -\frac{1}{L} \sum\nolimits_{i=1}^{L} \log \mathrm{P}\big(x_i \mid x_{0:i-1}\big) \tag{2}$$

and

$$\log \text{X-PPL}_{M_1,M_2} s = -\frac{1}{L} \sum\nolimits_{i=1}^{L} M_1 s_i \cdot \log\big(M_2 s_i\big) \tag{3}$$

The intuition is that human-written text diverges more from a language model's predictions than another language model does. By normalizing the observed perplexity against a baseline (the cross-perplexity), the method reduces sensitivity to prompt-induced variations. Texts with scores below a threshold are classified as machine-generated.

3.4 Out-of-distribution Detection using DeepSVDD

We evaluated the generalizability of the out-of-distribution (OOD) detection approach proposed by Zeng et al. [22]. OOD-DeepSVDD reframes machine-generated text detection as an out-of-distribution (OOD) detection problem. Rather than treating detection as binary classification between two classes, this approach treats machine-generated text as in-distribution (ID) samples and human-written text as out-of-distribution outliers.

The method uses DeepSVDD (Deep Support Vector Data Description), which learns a hypersphere in a high-dimensional embedding space that encapsulates the machine-generated training samples. A text encoder (e.g., RoBERTa) maps each input text x to an embedding $z = \phi x$. The model then learns a hypersphere center c and minimizes the training objective:

$$\mathcal{L}_{\mathrm{DSVDD}} = \frac{1}{N} \sum\nolimits_{i=1}^{N} \|\phi x_i - c\|^2 \tag{4}$$

The center c is computed as the mean of machine-generated sample embeddings, and training minimizes the distance of these samples from the center. At inference, the distance $\|\phi x - c\|$ is computed for each test sample. Texts with distances exceeding a threshold are classified as out-of-distribution (human-written), while those within the hypersphere are classified as machine-generated.

4 EXPERIMENTAL SETUP

4.1 Dataset

We utilize the EUR-LEX-57K dataset, which comprises 57,000 legislative documents in English from the EUR-Lex portal. From this dataset, we sampled 10,000 documents for synthetic generation, ensuring diversity across document types and lengths (average: 727 words). The dataset provides a challenging testbed for detection methods due to the formal, structured nature of legal language.

We generated synthetic legal documents using four state-of-the-art instruction-tuned language models Qwen3-8B, Llama-3.1-8B-Instruct, Mistral-7B-Instruct-v0.3, Gemma-3-4b-it. Each model generated 2,500 documents, yielding 10,000 synthetic samples total. All models operated in bfloat16 precision with batch size 32, maximum 2048 new tokens, temperature=0.7, and top-p=0.9. To create realistic paraphrases while maintaining legal authenticity, we employed two prompting strategies in equal proportion: (1) full anonymization with modified case numbers, and (2) case-number preservation using in-context learning with two-shot examples.

4.2 Detection Methods

We evaluate three zero-shot detection approaches:

BiScope: Computes bidirectional cross-entropy losses using GPT-2 and DistilGPT-2 as frozen feature extractors, extracting 20-dimensional feature vectors (5 statistics $\times$ 2 directions $\times$ 2 models) classified by Random Forest with 100 trees.

Binoculars: Compares perplexity measurements across Falcon-7B and Falcon-7B-Instruct models using the ratio of log-perplexity to cross-perplexity as a detection score.

OOD-DeepSVDD: Learns a hypersphere boundary around machine-generated embeddings using RoBERTa encoder.

4.3 Evaluation Metrics

We assess detection performance using four primary metrics: **Accuracy** (overall classification correctness), **Precision** (proportion of correctly identified AI texts among all texts flagged as AI), **Recall** (proportion of AI texts successfully detected), and **F1 Score**

5 RESULTS

We evaluate three detection methods (BiScope, Binoculars, and OOD-DeepSVDD) on 10,000 real EUR-Lex documents and 10,000 synthetic documents created from four generators.

5.1 Overall Performance

Table 1: Detection Performance (20,000 documents)

Method	Acc	Prec	Rec	F1
BiScope (no train)	50.1%	65.5%	0.6%	1.1%
BiScope (trained)	**74.5%**	**79.5%**	**66.0%**	**72.1%**
Binoculars	50.7%	50.4%	99.3%	66.8%
OOD-DeepSVDD	55.8%	61.0%	32.1%	42.1%

BiScope with training outperforms zero-shot methods but requires supervised learning—without training, it achieves only 50.1% accuracy with 0.6% recall (detecting 57 of 10,000 AI documents). Training improves accuracy by 24.4 points and AI detection 115-fold. Binoculars over-flags with 99.3% recall but 98% false positive rate. OOD-DeepSVDD misses 67.9% of AI text.

5.2 Generator and Length Effects

Table 2: BiScope Detection by Generator and Length

Generator	Overall	0-200w	200-400w	400-600w	600-800w
Qwen3-8B	86.0%	95.5%	85.4%	76.1%	78.6%
Gemma-3-4b	72.9%	92.9%	71.6%	64.7%	50.0%
Mistral-7B	55.2%	67.4%	58.7%	43.9%	27.3%
Llama-3.1-8B	51.1%	70.3%	47.9%	40.9%	38.1%
Overall	**66.0%**	**84.2%**	**65.5%**	**53.9%**	**50.6%**

Detection varies from 51%-86% across generators, with Llama-3.1 detecting at near-random rates (51.1%) while Qwen3-8B is reliably detected (86.0%) which is a 1.4× difference creating generator-specific bias. Performance degrades severely with length: 84.2% for short texts drops to 50.6% for 600-800 words. The 255 false positives (17%) mean one in six attorneys face unwarranted scrutiny, while 510 false negatives (34%) let substantial AI content pass undetected. Llama-3.1 and Mistral-7B account for 70% of detection failures (Appendix Table 3).

6 DISCUSSION

Our evaluation of three zero-shot detection methods on legal text reveals fundamental challenges when applying general-domain AI detectors to specialized domains.

BiScope exhibits severe limitations in the zero-shot setting, achieving only 50.1% accuracy with 0.6% recall, effectively random performance. The method flags 99.4% of AI-generated legal documents as human-written, detecting merely 57 of 10,000 AI samples. This failure stems from the inability of simple linear thresholds to separate overlapping bidirectional cross-entropy loss signals in formal legal language. However, training a Random Forest classifier on BiScope's features yields dramatic improvement: 74.5% accuracy with 66% recall, representing a 115-fold increase in detection capability. This demonstrates that while BiScope's bidirectional features capture meaningful signals, supervised learning is crucial for effectively exploiting them.

Despite this improvement, trained BiScope still exhibits critical weaknesses for deployment. Detection performance varies dramatically across generators (51.1% for Llama-3.1 to 86.0% for Qwen3-8B), creating a 1.4× bias that raises fairness concerns and users of certain models face substantially different detection risks despite identical use cases. Performance also degrades severely with document length: 84.2% recall for documents under 200 words drops to 50.6% for 600–800 word documents, directly contradicting legal practice where documents typically exceed 400 words. The error distribution carries professional consequences: 255 false positives (17%) subject innocent attorneys to unwarranted scrutiny, while 510 false negatives (34%) allow substantial AI content to pass undetected.

Binoculars falls into the opposite trap: 99.3% recall but 98% false positive rate. The method interprets the low perplexity inherent in rigid, repetitive legal language as evidence of AI-generated content, incorrectly flagging nearly all human-written legal documents as machine-generated. This demonstrates that perplexity-based approaches designed for diverse natural languages fail when applied to constrained, formal domains.

OOD-DeepSVDD achieves only 55.8% accuracy with 32.1% recall, missing 67.9% of AI-generated text. The method's core assumption that machine-generated text forms a compact, learnable cluster separate from human text breaks down when synthetic data is of high quality and domain-authentic. Our case-number preservation strategy and legal-specific prompting produced synthetic documents that are so realistic they do not form a distinct distributional cluster, thereby undermining the OOD detection paradigm.

These findings reveal a fundamental tension: effective detection requires domain-specific training data and continuous adaptation as models evolve, while practical deployment demands zero-shot generalization to unknown models and domains. The legal domain amplifies these challenges through its formal structure, specialized vocabulary, and citation patterns that confound features designed for general text.

7 CONCLUSION

This work evaluates three detection methods (BiScope, Binoculars, and OOD-DeepSVDD) for identifying AI-generated legal text across 20,000 real and synthetic EUR-Lex documents from four generators. In this realistic legal setting, none of the methods achieves reliability suitable for real-world deployment.

BiScope, when trained with domain-specific labels, performs best with 74.5% accuracy and 66% recall, but exhibits strong generator-specific bias, length-dependent degradation on 400–800 word documents, and substantial error rates (17% false positives, 34% false negatives). Binoculars attains 99.3% recall but at the cost of an extreme false positive rate, while OOD-DeepSVDD remains only slightly better than random chance, indicating that current zero-shot and OOD-style detectors are not robust enough.

These findings highlight that effective legal-domain detection requires training and ensemble methods that incorporate legal-specific signals rather than relying on a single, static "zero-shot" detector.

Our findings suggest three directions for future work: (1) domain adaptation through fine-tuning surrogate models on legal corpora, (2) incorporating legal-specific signals (citation patterns, statutory structure), and (3) ensemble approaches combining multiple detection methods.

8 PROJECT CONTRIBUTIONS

- **Abhilash Shankarampeta:** Worked on Binoculars and OODD-SVDD experimentation and results gathering. Wrote the methodology and setup section.

- **Hemanth Bodala:** Worked on Biscope experimentation and results gathering. Wrote results and discussion section.
- **Vignesh Palaniappan:** Worked on synthetic data generation + validation. Wrote introduction + conculsion section and did overall report review.
- **Rohan Surana:** Worked on synthetic data generation + prompt development. Wrote synthetic data generation section.
- **Ananay Gupta:** Worked on synthetic data generation. Wrote abstract section and did overall report review. Presented in class
- **Sarthak Kala:** Presented the project in the class, did a review of the final report, and conducted some experiments based on a related paper that resulted in negative findings.

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A APPENDIX

A.1 Synthetic Data Generation Prompts

Prompt Variant 1 (preserve_case_numbers=False):

Synthetic Generation Prompt - Full Anonymization

```
You are tasked with creating a synthetic version
of a legal document text. Rewrite and paraphrase
the following legal text while:
1. Changing all dates, years, reference numbers,
and case numbers
2. Changing any person names, organization names,
or location names (PII)
3. Changing specific details (e.g., diseases,
countries, rivers, etc.) while keeping the legal
structure
4. Maintaining the formal legal language style
5. Keeping similar length and structure
Original Text: {text}
Provide ONLY the rewritten text, without any
explanation or additional text.
```

Prompt Variant 2 (preserve_case_numbers=True):

Synthetic Generation Prompt - Preserve Case Numbers

```
Task: Rewrite legal documents while keeping case
numbers identical.
Example 1: Original: "Case T-123/15 was decided
on 12 January 2016 in Brussels." Rewritten:
"Case T-123/15 was resolved on 5 March 2018 in
Luxembourg."
Example 2: Original: "In judgment C-456/19, the
applicant Smith from Italy appealed." Rewritten:
"In judgment C-456/19, the appellant Johnson from
Spain contested."
Your turn: Original: {text}
Rewritten:
```

A2. Detailed Performance Metrics

A3. Detailed Length Analysis

A4. Complete Cross-Tabulation

A5. Error Characteristics

False Negative Distribution by Generator:

- Llama-3.1-8B: 186 (48.95% of Llama samples)
- Mistral-7B: 169 (44.83% of Mistral samples)
- Gemma-3-4b-it: 105 (27.13% of Gemma samples)
- Qwen3-8B: 50 (14.04% of Qwen samples)

False Positive Distribution by Length:

- 0-200 words: 74 (29.0%)
- 200-400 words: 107 (42.0%)
- 400-600 words: 38 (14.9%)
- 600-800 words: 12 (4.7%)
- 800-1000 words: 4 (1.6%)
- 1000+ words: 20 (7.8%)

A6. Confidence Calibration

A7. Training, Validation, and Test Performance

A8. Training Necessity Analysis

Without training, the model predicts 99.56% of documents as human-written (19,913 of 20,000), detecting only 57 AI documents. This represents catastrophic failure: 99.43% false negative rate with near-zero utility. Training improves AI detection 115-fold (0.57% to 66.00%), demonstrating supervised learning is essential for BiScope functionality.

Table 3: Extended Performance Metrics by Generator

Generator	Samples	Correct	Wrong	Detection	FN Rate
Qwen3-8B	356	306	50	85.96%	14.04%
Gemma-3-4b-it	387	282	105	72.87%	27.13%
Mistral-7B	377	208	169	55.17%	44.83%
Llama-3.1-8B	380	194	186	51.05%	48.95%
Total	**1,500**	**990**	**510**	**66.00%**	**34.00%**

Table 4: Performance Stratified by Document Length

Length (words)	Total Docs	Human Count	AI Count	Overall Acc	Human Acc	AI Detect
0-200	601	310	291	80.03%	76.13%	84.19%
200-400	1,585	782	803	75.77%	86.32%	65.50%
400-600	492	258	234	70.33%	85.27%	53.85%
600-800	162	75	87	66.05%	84.00%	50.57%
800-1000	70	28	42	62.86%	85.71%	47.62%
1000+	90	47	43	62.22%	57.45%	67.44%

Table 5: Detection Rate by Generator and Length (Complete)

Generator	0-200	200-400	400-600	600-800	800-1000	1000+
Qwen3-8B	95.51%	85.39%	76.09%	78.57%	77.78%	83.33%
Gemma-3-4b-it	92.94%	71.63%	64.71%	50.00%	30.77%	57.14%
Llama-3.1-8B	70.27%	47.92%	40.85%	38.10%	44.44%	69.23%
Mistral-7B	67.44%	58.72%	43.94%	27.27%	45.45%	64.71%

Table 6: False Positive and False Negative Analysis

Metric	False Positives	False Negatives
Total Count	255	510
Rate	17.00%	34.00%
Mean Word Count	405	405
Median Word Count	285	342
Avg Confidence	0.591	0.397

Table 7: Prediction Accuracy by Confidence Level

Confidence	Predictions	Accuracy	Coverage	% of Total
Very Low (0-0.3)	496	87.50%	496	16.53%
Low (0.3-0.5)	1,259	64.42%	1,259	41.97%
Medium (0.5-0.7)	747	68.81%	747	24.90%
High (0.7-0.9)	347	93.66%	347	11.57%
Very High (0.9-1.0)	151	100.00%	151	5.03%

Table 8: Performance Across Data Splits

Split	Samples	Accuracy	F1-Score	AUROC
Training	14,008	85.92%	-	-
Validation	2,992	74.23%	0.7230	0.8305
Test	3,000	74.50%	0.7213	0.8256
Train-Test Gap	-	**11.42%**	-	-

Table 9: Impact of Supervised Training

Approach	Acc	Prec	Rec	F1	FP Rate	FN Rate
No training	50.13%	65.52%	0.57%	1.13%	0.30%	99.43%
Trained RF	74.50%	79.52%	66.00%	72.13%	17.00%	34.00%
Gain	**+24.37%**	**+14.00%**	**+65.43%**	**+71.00%**	-	**-65.43%**